The Management University of Africa

MUA

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**BDS 109 CAT** **ANSWER ALL QUESTIONS**

1. 250 members of a certain society have voted to elect a new chairman. Each member may vote for either one or two candidates. The candidate elected is the one who polls most votes. Three candidates x, y z stood for election and when the votes were counted, it was found that

   59 voted for y only, 37 voted for z only

   12 voted for x and y, 14 voted for x and z

   147 voted for either x or y or both x and y but not for z

   102 voted for y or z or both but not for x

   **Required**

   i. Present the above information in a Venn diagram. **(6 Marks)**

   ii. How many voters did not vote? **(1 Mark)**

   iii. How many voters voted for x only? **(1 Marks)**

   iv. Who won the elections? **(1 Mark)**

2. The weekly wage of 2,000 workmen is normally distributed with a mean wage of Shs. 70 and wage standard deviation of Shs. 5. Estimate the number of workers whose weekly wages are:

   i. More than Shs. 72 **(3 marks)**

   ii. Less than Shs. 65 **(3 marks)**